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CANDIDATE
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CHEMISTRY

Grade 12

Paper 3

May 2022

2 hours

Candidates answer on the Question Paper.

Additional Materials: Calculator
Qualitative Analysis Notes and Periodic Table
Ruler

12CHEM/03

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You may receive fewer marks if you do not show your working or do not use appropriate units where required.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 40.

Answer all questions in English.

For Examiner's Use	
1	
2	
Total	

This document consists of **11** printed pages and **1** blank page.



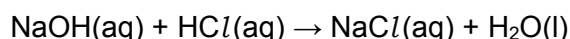
Read all the instructions for Question 1 carefully before starting the experiment.

For
Examiner's
Use

- 1 In this experiment you will find the percentage purity of calcium carbonate in a sample of crushed limestone. You will first react a known mass of the crushed limestone with a known excess amount of hydrochloric acid, $\text{HCl}(\text{aq})$, to make a solution. You may assume that only the calcium carbonate present in the sample reacts with the acid.



The amount of acid that did not react with the carbonate is then found by titration using aqueous sodium hydroxide, $\text{NaOH}(\text{aq})$. You may assume that the impurities in the limestone do **not** react with the sodium hydroxide.



FB 1 is crushed limestone that contains calcium carbonate and a soluble impurity.

FB 2 is $2.00 \text{ mol dm}^{-3} \text{ HCl}$.

FB 4 is $0.200 \text{ mol dm}^{-3} \text{ NaOH}$

phenolphthalein

In calculations use values with three significant figures.

(a) (i) Determining the mass of the FB 1

- Measure and record the mass of the container and **FB 1** in a table.
- Transfer all the sample of **FB 1** into a 250 cm^3 beaker.
- Measure and record the mass of the empty container in the table.
- Calculate and record the mass of the sample **FB 1** in the table.

[2]

(ii) Making the solution

- Pipette 50.0 cm^3 of **FB 2** and transfer this solution into the beaker containing **FB 1**.
- Stir the mixture carefully in the beaker until all the sample has dissolved.
- Transfer the contents of the beaker into the 250 cm^3 volumetric flask.
- Rinse the beaker several times with distilled water and transfer all the contents to the volumetric flask.
- Make this solution up to the mark on the volumetric flask with distilled water and mix thoroughly by shaking the flask. Label this solution **FB 3**.

(b) Titration

- Fill a burette with **FB 3** from the volumetric flask.
- Pipette 25.0 cm³ of **FB 4** into a conical flask.
- Add a few drops of phenolphthalein indicator to the conical flask.
- Do a rough titration and record your burette readings in the table.
- Do as many accurate titrations as you think necessary to obtain consistent results.
- Record, in the table, all of your burette readings and the titre (the volume of **FB 3** added).

Repeat until you obtain concordant results.

- (i) Record the results from the titrations in this table. You **do not** need to use all the columns.

titration number					
initial (burette) reading / cm³					
final (burette) reading / cm³					
titre / cm³					

[7]

- (ii) Calculate the mean titre.

State which titrations you use to calculate the mean titre.

titrations used

mean titre cm³ [2]

(c) Calculations

Show your working and quote the final answer to each step of your calculations to an appropriate number of significant figures.

- (i) Calculate the number of moles of sodium hydroxide present in 25.0 cm³ of **FB 4**.

moles of NaOH = mol [1]

- (ii) Hence calculate the number of moles of hydrochloric acid present in the volume of **FB 3** calculated in (b)(ii).

For
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Use

moles of HCl = mol [1]

- (iii) Use your answer from (c)(ii) to calculate the number of moles of hydrochloric acid present in 250 cm^3 of **FB 3**.

moles of HCl in 250 cm^3 of **FB 3** = mol [1]

- (iv) Calculate the number of moles of hydrochloric acid, **FB 2**, added to **FB 1** in (a).

moles of HCl added to **FB 1** = mol [2]

- (v) Use your answers to (c)(iii) and (c)(iv) to calculate the number of moles of hydrochloric acid that reacted with the calcium carbonate in **FB 1**.

For
Examiner's
Use

moles of HCl reacted with CaCO_3 = mol [1]

- (vi) Calculate the number of moles of calcium carbonate present in your sample of **FB 1**.

moles of CaCO_3 = mol [1]

- (vii) From your answer to (c)(vi) and the mass of **FB 1** used in (a), calculate the percentage purity by mass of calcium carbonate in the limestone.

[Ar : C, 12.0; O, 16.0; Ca, 40.1]

percentage purity of calcium carbonate = % [2]

[Total: 20]

Read all the instructions for Question 2 carefully before starting the experiment.

For
Examiner's
Use

2 Qualitative Analysis

At each stage of any test you are to record details of the following.

- colour changes seen
- the formation of any precipitates
- the solubility of such precipitates in an excess of the reagent added

Where gases are released they should be identified by a test, **described in the appropriate place in your observations.**

You should indicate clearly at what stage in a test a change occurs.

Marks are **not** given for chemical equations.

No additional tests for ions present should be attempted.

If any solution is warmed, a boiling tube MUST be used.

Rinse and reuse test-tubes and boiling tubes where it is possible.

Where reagents are selected for use in a test, the name or correct formula of the element or compound must be given.

(a) You will do tests to identify the salt **FB 5**.

FB 5 is an aqueous solution of a salt.

It contains one cation, which is listed in the Qualitative Analysis Notes.

It also contains one anion, which is either sulfate, SO_4^{2-} , or sulfite, SO_3^{2-} .

(i) Plan tests to identify the cation in **FB 5**. Write the details of each test in the table. You do not need to complete each row in the table.

Do these tests **and** record your observations in the Table 2.1.

Table 2.1

test	observation

The cation in **FB 5** is

[5]

- (ii) Plan one of the tests to identify if the anion in **FB 5** is sulfate or sulfite. Write the details of the test in the table. You do not need to complete each row in the table.

Do the test and record your observations in the Table 2.2.

Table 2.2

test	observation

The anion in **FB 5** is

[3]

For
Examiner's
Use

(b) **FB 6**, **FB 7** and **FB 8** are aqueous solutions of salts.

- Each salt contains one cation and one anion.
- One of the ions is sodium; the other five ions are listed in the Qualitative Analysis Notes.
- Two salts have a different nitrogen-containing ion.
- **FB 7** or **FB 8** contains a halide ion.

(i) You will identify the cations present in **FB 6**, **FB 7** and **FB 8**.

You will do six separate tests.

- Add 2 cm depth of dilute sulfuric acid to 1 cm depth samples of **FB 6**, **FB 7** and **FB 8** in separate test tubes.
- Add aqueous sodium hydroxide drop by drop until in excess to 1 cm depth samples of **FB 6**, **FB 7** and **FB 8** in separate boiling tubes.

Record all your observations in a suitable table in the space below.

[4]

- (ii) Suggest the tests and tests results for gases formed in **(b)(i)** and fill the table below.

You do **not** need to complete each row in the Table 2.3.

Table 2.3

gas	test results

[2]

- (iii) **FB 7** or **FB 8** contains a halide ion.

Describe the test you will use to identify the halide ion present in either **FB 7** or **FB 8**.

Test **FB 7** and **FB 8** for halide ions record your observations in the Table 2.4 below.

test used

Table 2.4

solution	observation	halide ion present
FB 7		
FB 8		

[2]

(iv) **FB 8** contains a nitrogen-containing ion.

Describe the test you will use to confirm that **FB 8** contains a nitrogen-containing ion.

Test **FB 8** to confirm the presence of a nitrogen-containing ion and record your observations in the Table 2.5 below.

Table 2.5

test used	observation

[2]

(v) Use the information given in (b) and your observations in all tests to deduce the chemical formulae of the three salts.

FB 6 is **FB 7** is **FB 8** is

[2]

[Total: 20]

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